LunchLens

See tomorrow's lunch crowd before the soup hits the pot.

Menu-aware hybrid forecasting for campus lunch demand and food-waste reduction.

Aalto AI Open Source Lab · Hybrid ML + LLM · **MVP live**

Run the MVP → · Download PDF deck →

Speaker notes: Open with the project name and tagline — let it land. One breath: hybrid AI for campus lunch forecasting and food-waste reduction. Frame it as applied research, not a weekend demo.

10:30 AM. Prep starts for 200 lunches.

Signal	Today
Weather	Pouring rain
Competition	Free guild pizza nearby
Menu	Liver casserole special

By 2 PM: Half the trays come back untouched — not carelessness, but **invisible demand**.

Speaker notes: Paint the scene slowly — rain, guild pizza, liver casserole. Pause before the punchline. Key line: demand was invisible until it was too late.

Campus kitchens cook blind

Lunch spots prep on instinct. Nearby prices are roughly equal — so **something else drives the crowd**.

Over-prep

→ food in the bin

Under-prep

→ stockouts and lost revenue

Speaker notes: Kitchens prep on gut feel — everyone on campus knows this rhythm. Prices are similar, so price is not the lever. Two failure modes: waste vs stockouts.

What actually moves the needle

Signal	Why it matters
Today's menu	Not every dish draws the same crowd
Weather	Rain and cold change walk-in behavior
Day of week	Tuesday lunch is not Friday lunch
Competitor menus	Guild pizza next door pulls demand away

Speaker notes: Walk through the four signals. Use a campus example: guild pizza day steals your lunch rush. Transition: both bad outcomes share one root cause — guessing demand.

Two bad outcomes, one root cause

Over-prepare

Avoidable waste — cost, climate, full trays returned

Under-prepare

Stockouts, lost revenue, peak-hour chaos

Both come from guessing demand instead of forecasting it.

Speaker notes: Over-prepare hits cost, climate, and morale. Under-prepare means angry customers and a scrambling kitchen. Land the line: both come from forecasting by guesswork.

Our research question

> *We are asking a research question with a real sustainability payoff.*

RQ1 Does hybrid ML + LLM beat classical forecasting?

RQ2 How much waste and cost disappear vs a fixed baseline prep policy?

RQ3 Weather, own menu, or competitors — what matters most?

Speaker notes: Pivot tone: this is not another dashboard. Read RQ1–RQ3 clearly; judges care about depth. RQ2 is the sustainability hook.

LunchLens in one sentence

A hybrid forecaster that reads the menu, checks the weather, scans competitor menus, and estimates how many people are coming.

ML	LLM	Output
Numbers from history	Context from menus	One unified forecast

Less guesswork. Less waste. More lunch.

Speaker notes: Read the one-liner verbatim — it is the elevator pitch. Three beats: ML for numbers, LLM for menu context, one fused forecast.

Step 1 — Look back

Tabular baselines on history and calendar

Historical counts capture weekday rhythms, holidays, and semester patterns.

Baseline	Method
Naive weekday mean	Historical average by day of week
Linear regression	Calendar and date features
Gradient boosting	XGBoost / LightGBM on tabular signals

Speaker notes: Start the three-step pipeline: look back, look around, look ahead. This is the floor we need to beat before adding LLM signals.

Step 2 — Look around

LLM scores demand signals from 1 to 5

The LLM **rates** menu appeal, competitor pressure, and weather pull — it does not invent visitor counts.

Input	Role
Date	Calendar and semester context
Own menu	Semantic appeal of today's dishes
Competitor menus	Local competition pressure
Weather	Walk-in sensitivity to rain and temperature

Speaker notes: Critical distinction: the LLM does not invent visitor counts. Structured JSON output — not free-text guessing.

Step 3 — Look ahead

Hybrid fusion → operational outputs

Output 1

Estimated daily visitors

Fused tabular + LLM forecast → recommended prep

Output 2

Baseline vs model cost (€)

70% max-capacity baseline · 10 €/kg waste · 5 € shortage penalty

Cumulative cost charts show the running difference over time

Speaker notes: Fusion combines tabular ML and LLM scores. Compare against a simple baseline: prep for 70% of max capacity. Track waste kg, shortage penalties, and cumulative € saved.

Structured LLM output

JSON scores with guardrails

```json
{
  "menu_appeal": 4,
  "competitor_pressure": 2,
  "weather_effect": 3,
  "overall_demand_signal": 3.5
}
```

JSON schema · scores clamped 1–5 · prompts logged for reproducibility

Speaker notes: Show the JSON — judges want structured outputs, not vibes. Emphasize reproducibility — every score is cached and auditable.

Ablation study

What does each signal buy us?

Model	Signals	Expected lift
Naive	Weekday average	Baseline to beat
+ Weather	Rain and temperature	Small gain
+ LLM menu	Semantic appeal	Does text help?
+ Competitors	Local competition	Full context
LunchLens full	All fused	Best forecast, waste, and cost savings

Metrics: MAE · waste saved % · daily and **cumulative cost difference (€)**

Speaker notes: This table is the research flex — incremental signal additions. Hypothesis: full LunchLens fusion wins on both forecast and waste.

Who it is for

Starts on campus. Does not stay there.

Audience	Why they care
Campus cafés	Less waste, fewer stockouts
Sustainability teams	Reportable waste-saved metric
ML / NLP researchers	Hybrid benchmark: menus + time series
Future OSL teams	Fork, plug in real data, rerun eval

Speaker notes: Campus cafés get operational value. Sustainability teams get a metric they can report. Researchers get an open benchmark.

What we shipped

Layer	Deliverable
Data	730-day synthetic campus dataset (menus, weather, competitors)
Backend	FastAPI — hybrid forecast, LLM scorer, waste/cost API
Frontend	React dashboard — visitor chart, waste kg, cumulative cost (€)
Baseline	Fixed prep at **70% of max capacity** vs LunchLens model

Run locally: `backend` + `frontend` · OpenAI key for live LLM scoring · see MVP.md

Speaker notes: MVP is runnable today — not a slide-deck prototype. Show the dashboard: pick a date, see baseline vs model prep, waste in kg, and cumulative € difference. Synthetic data now; partner restaurant pilot next.

Why it matters

Food waste is not abstract. It is trays coming back full while someone else goes hungry.

Tech for Good — turn forecast error into waste kg and € people can feel.

Open source — benchmark, eval harness, prompt templates, live MVP — fork and extend.

We go beyond courses. We build, we measure, we share.

Speaker notes: Slow down for the close. Final line: We go beyond courses. We build, we measure, we share. Pause. Thank the judges. Offer to walk through the ablation design.